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Branched-chain amino acids for people with cirrhosis and hepatic encephalopathy (Review)

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[Intervention Review]

Branched-chain amino acids for people with cirrhosis and hepatic encephalopathy

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ABSTRACT

Rationale

Hepatic encephalopathy is a brain dysfunction characterised by neurological and psychiatric changes due to liver insufficiency or portal-systemic shunting. The severity ranges from minor symptoms to coma. The previous version of this updated review meta-analysed data from 16 randomised clinical trials on branched-chain amino acids (BCAAs) versus control interventions, and found that BCAAs did not affect mortality but had a beneficial effect on hepatic encephalopathy. As data on critical outcomes were insufficient and new trials were published, we updated the review again.

Objectives

To assess the beneficial and harmful effects of BCAAs versus any control intervention for people with cirrhosis and hepatic encephalopathy.

Search methods

We identified trials through manual and electronic searches in the Cochrane Hepato-Biliary Group Controlled Trials Register, the Cochrane Central Register of Controlled Trials, MEDLINE, Embase, Latin American and Caribbean Health Science Information database (LILACS), Science Citation Index Expanded, and Conference Proceedings Citation Index – Science. We also searched other resources and contacted experts for additional published and unpublished trials. The latest search date was 12 December 2024.

Eligibility criteria

We included randomised clinical trials, irrespective of bias control, language, outcomes reported, or publication status, that compared any form of branched-chain amino acid with no intervention, placebo, diets, non-absorbable disaccharides, antibiotics, or any other intervention with a potential effect on hepatic encephalopathy, in children and adults with overt or minimal hepatic encephalopathy associated with acute or chronic liver disease.

Outcomes

The critical outcomes were all-cause mortality, hepatic encephalopathy (number of people without improved manifestations of hepatic encephalopathy), and serious adverse events (including nausea and diarrhoea). The important outcomes were quality of life and markers of nutritional status, including serum albumin and nitrogen balance. We evaluated outcomes at the longest available follow-up duration. The longest follow-up was also our primary time point for analysis.

Risk of bias

We assessed the risk of bias (RoB) of the critical and important outcomes using the Cochrane RoB 2 tool.

Synthesis methods

We used standard Cochrane methodology. We performed meta-analyses, based on intention-to-treat, with risk ratios (RRs) for dichotomous outcomes, standardised mean differences (SMDs) for continuous outcomes in trials using different scales, and mean differences (MDs) for continuous outcomes when trials used the same scales, all with 95% confidence intervals (CI). We conducted the main analysis using a random-effects model. We assessed the overall certainty of the evidence per outcome with the GRADE approach, with five components (risk of bias, indirectness, heterogeneity, imprecision using the minimally contextualised approach, and dissemination bias).

Included studies

We added two new randomised clinical trials to this review update. Thus, the 18 trials, published between 1984 and 2023, included 934 adults (mean age 47 to 64 years). We found no trials in children. Participants had overt hepatic encephalopathy (13 trials) and covert hepatic encephalopathy (five trials). Ten trials assessed oral BCAA supplements and eight trials assessed intravenous BCAAs. The control groups received no intervention or placebo (two trials), diets (11 trials), lactulose (three trials), or neomycin (two trials). Most participants had cirrhosis (minimum 97%). One trial included participants with acute hepatitis. The follow-up periods were from four days to two years (104 weeks). Ten trials were conducted in Europe, four in Asia, two in the USA, and one each in Brazil and Australia. Three trials received support from for-profit organisations in the form of interventions.

Synthesis of results

BCAAs may have little to no effect on all-cause mortality compared with the control interventions, but the evidence is very uncertain (RR 0.89, 95% CI 0.71 to 1.12; 17 studies, 867 participants; very low-certainty evidence). We found no evidence of small-study effects. All trials reported the effect of BCAAs on hepatic encephalopathy. The evidence suggests that BCAAs reduce hepatic encephalopathy (RR 0.79, 95% CI 0.64 to 0.96; 18 studies, 934 participants; low-certainty evidence). The follow-up ranges for the two outcomes were four days to two years.

Nausea and diarrhoea, considered as serious adverse events, occurred in 58 of 477 (12%) participants in the BCAAs group and 16 of 538 (3%) participants in the control intervention group (diets). The evidence is very uncertain about the effect of BCAAs on nausea and diarrhoea (RR 2.05, 95% CI 0.40 to 10.58; 6 studies, 1015 participants; very low-certainty evidence).

Three trials provided data on quality of life, but we were unable to meta-analyse the data because of the methods used to register the scores. All participants in the three trials had cirrhosis, but some did not have hepatic encephalopathy at baseline. When the analyses were limited to people with hepatic encephalopathy, none of the trials found beneficial or harmful effects of BCAAs on the global 36-item Short Form Health Survey (SF-36) score or any of the subscales. BCAAs may have little to no effect on albumin concentration (range 12 to 56 weeks) (MD 0.60, 95% CI -0.90 to 2.09; $I^2 = 0\%$; 3 studies, 176 participants; very low-certainty evidence), but the evidence is uncertain. The evidence is very uncertain about the effect of BCAAs on nitrogen balance (range four days to two years) (SMD 0.82, 95% CI -1.01 to 2.64; 3 studies, 108 participants; very low-certainty evidence).

We downgraded the certainty of evidence for risk of bias, indirectness, and imprecision.

Three trials are ongoing.

Authors' conclusions

We added two new trials to the analyses. The evidence suggests BCAAs reduce hepatic encephalopathy, but the certainty of evidence is low. We do not know if BCAAs, compared with controls, have any effect on all-cause mortality, nausea and diarrhoea, albumin, and nitrogen balance because of very low-certainty evidence. We could not meta-analyse the data on quality of life. Trials in children are lacking. We lack randomised clinical trials comparing BCAAs with interventions such as non-absorbable disaccharides, rifaximin, or other antibiotics.

Funding

No funding

Registration

Protocol (1997): Gluud C, Koretz RL. Branched-chain amino acids for hepatic encephalopathy (Protocol for a Cochrane Review). The Cochrane Library 1997, Issue 1.

Original review (2003): doi.org/10.1002/14651858.CD001939

Review update (2015 Feb): doi.org/10.1002/14651858.CD001939.pub2

Review update (2015 Sep) doi.org/10.1002/14651858.CD001939.pub3

Review update (2017): doi.org/10.1002/14651858.CD001939.pub4

PLAIN LANGUAGE SUMMARY

Do branched-chain amino acids (BCAAs) help people with cirrhosis and hepatic encephalopathy (a brain disorder caused by severe liver disease)?

Key messages:

- BCAAs may have little to no effect on the number of people with hepatic encephalopathy who die, compared to other treatments or no treatment, but we are very uncertain about the results.
- BCAAs may reduce the number of people who develop hepatic encephalopathy.
- We need more high-quality studies that give participants BCAAs combined with other medicines used in practice, such as antibiotics.

What are cirrhosis and hepatic encephalopathy?

Cirrhosis is a condition in which healthy liver tissue is gradually replaced by scar tissue. This damage reduces the liver's ability to work properly and is the main cause of hepatic encephalopathy, a condition that affects brain function.

Symptoms can range from mild to severe. Mild symptoms include small, almost unnoticeable, changes in thinking, such as problems with attention, memory, or problem-solving abilities. In more severe stages of the condition, symptoms are clearly noticeable. These may include obvious confusion or disorientation, changes in behaviour or personality, or reduced alertness. In very severe cases, people may lose consciousness or enter a coma.

How is hepatic encephalopathy treated?

Most treatments for hepatic encephalopathy aim to lower toxins in the blood, especially ammonia, and to manage the underlying liver disease. Doctors often use medicines that reduce the amount of toxins produced in the gut or stop them from being absorbed into the bloodstream.

Branched-chain amino acids (BCAAs) may also be used as part of treatment, but they work in a different way. People with cirrhosis often have low levels of BCAAs. Taking BCAA supplements can help correct this imbalance and support muscles in removing ammonia from the blood, which may help improve brain function.

BCAAs are amino acids with a specific chemical structure and are important for maintaining muscle mass and supporting normal brain signalling. Because of these roles, BCAA supplementation may help people with hepatic encephalopathy.

What did we want to find out?

We wanted to know if BCAAs are an effective and safe treatment for people with hepatic encephalopathy and cirrhosis. We were interested to know if they made any difference to these outcomes: death, development or progression of hepatic encephalopathy, serious unwanted or harmful events (also known as 'adverse events'), quality of life, and markers of nutritional status.

What did we do?

We searched for studies that compared BCAAs to another treatment, including placebo (an inactive or 'dummy' medication), no intervention, diets, lactulose (a liquid sugar often used to treat constipation), or neomycin (an antibiotic) in people with hepatic encephalopathy and cirrhosis.

We compared and summarised the results of the studies and rated our confidence in the evidence, based on factors such as study methods and size.

What did we find?

We found 18 studies involving 934 adults. The participants had cirrhosis, often due to alcoholic liver disease or viral hepatitis (liver infection due to a virus). Participants' average age ranged from 47 to 64 years. The proportion of men in the studies ranged from 47% to 90%. The studies' follow-up durations ranged from four days to two years. Ten trials were conducted in Europe, four in Asia, two in the USA, and one each in Brazil and Australia. We found no studies involving children.

Main results

Our analyses demonstrated that BCAAs:

- may make little to no difference to the number of people who die, but the evidence is very uncertain;

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- may reduce the number of people who develop hepatic encephalopathy or whose condition worsens;
- may increase the risk of nausea and diarrhoea, but the evidence is very uncertain.

The evidence regarding the effect of BCAAs on markers of nutritional status is very uncertain. We were unable to analyse evidence on people's quality of life due to the way studies assessed this outcome, and because some studies included participants without hepatic encephalopathy.

What are the limitations of the evidence?

We are not confident in the evidence because many studies did not provide sufficient information to adequately assess how well they were designed or conducted. We also had concerns about the possibility of selective reporting, which is when a study shares only some of its results, which can give a misleading picture of the true effect of treatment. In addition, most studies were small.

How current is this evidence?

This review updates our previous review from 2017. The evidence is current to 12 December 2024.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Branched-chain amino acids (BCAAs) for people with cirrhosis and hepatic encephalopathy

Branched-chain amino acids (BCAAs) for people with cirrhosis and hepatic encephalopathy

Patient or population: hepatic encephalopathy

Setting: hospitalised and outpatient

Intervention: BCAAs

Comparison: control

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with control	Risk with BCAAs				
All-cause mortality follow-up: range 4 days to 2 years	256 per 1000	228 per 1000 (179 to 287)	RR 0.89 (0.70 to 1.12)	867 (17 RCTs)	⊕⊕⊕⊕ Very low ^{a,b}	With control, 256 out of 1000 people would die, and 744 would not. With BCAAs, 228 out of 1000 people would die, and 772 would not. BCAAs may have little to no effect compared with the control interventions. The evidence is very uncertain.
Hepatic encephalopathy follow-up: range 4 days to 2 years	592 per 1000	468 per 1000 (379 to 568)	RR 0.79 (0.64 to 0.96)	934 (18 RCTs)	⊕⊕⊕⊕ Low ^a	With control, 592 out of 1000 people would develop hepatic encephalopathy, and 408 would not. With BCAAs, 468 out of 1000 people would develop hepatic encephalopathy, and 532 would not. The evidence suggests that BCAAs reduce hepatic encephalopathy. The evidence is uncertain.
Serious adverse events (nausea and diarrhoea) follow-up: range 7 days to 2 years	30 per 1000	61 per 1000 (12 to 315)	RR 2.05 (0.40 to 10.58)	1015 (6 RCTs)	⊕⊕⊕⊕ Very low ^{a,c}	With control, 30 out of 1000 people would develop nausea or diarrhoea, and 970 would not. With BCAAs, 61 out of 1000 people would develop nausea or diarrhoea, and 939 would not. The evidence is very uncertain. Deaths are not counted in the analyses of serious adverse events (SAEs). One trial reported SAEs other than nausea and diarrhoea (Mehtani 2023).
Quality of life	We were unable to perform a meta-analysis as the studies used different scoring and reporting methods for the 36-item Short Form Health Survey (SF-36) instrument.			(3 RCTs)	-	All participants in the trials had cirrhosis, but some did not have hepatic encephalopathy at baseline. When the analysis was limited to people with hepatic encephalopathy, none of the trials found beneficial or harmful effects of BCAAs on quality of life.

Serum albumin follow-up: range 12 weeks to 56 weeks	The mean serum albumin ranged from 29 to 34.3 g/L	MD 0.6 g/L higher (0.9 lower to 2.09 higher)	-	176 (3 RCTs)	⊕⊕⊕⊕ Very low ^{d,e,f}	With control, the serum albumin would be 29 g/L. With BCAAs, the serum albumin would be 29.60 g/L. The evidence regarding the effect of BCAAs on serum albumin is very uncertain.
Nitrogen balance follow-up: range 4 days to 2 years	-	SMD 0.82 SD higher (1.01 lower to 2.64 higher)	-	108 (3 RCTs)	⊕⊕⊕⊕ Very low ^{d,e,f,g}	Average difference (SMD): 0.82 SD higher (95% CI 1.01 SD lower to 2.64 SD higher), based on data from 108 participants in 3 trials. The evidence regarding the effect of BCAAs on nitrogen balance is very uncertain. A fourth trial measured nitrogen balance as an outcome (Calvey 1985). No difference in nitrogen balance was demonstrated between the BCAA and treatment groups. The results could not be included in the meta-analysis, as some participants also had non-cirrhotic liver disease.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **MD:** mean difference; **RR:** risk ratio; **SMD:** standardised mean difference

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradeapro.org/presentations/#/isof/isof_question_revman_web_447899099654717676.

^a Downgraded two levels for risk of bias because most of the included trials were judged to be at high risk of bias or raised some concerns.

^b Downgraded one level due to imprecision as the CI crosses the null effect, including important benefit and unimportant harm.

^c Downgraded two levels due to imprecision as the CI crosses the null effect, including important benefit and important harm.

^d Downgraded one level for indirectness (indirect outcome used as a surrogate estimate of nutritional status).

^e Downgraded two levels for imprecision in the overall assessment (the CI crosses the null effect, and the sample size does not meet the optimal information size (OIS) of 400 participants).

^f Due to the small number of trials, we were unable to determine the risk of publication bias.

^g Downgraded one level for risk of bias (we judged two of the three trials to have some concerns for bias in the selection of reported results).

BACKGROUND

Description of the condition

Cirrhosis is a late stage of scarring (fibrosis) of the liver, caused by many types of liver disease and conditions. Each time the liver is injured — after excessive alcohol consumption, by disease, overweight, or another cause — it tries to repair itself. In the process, scar tissue forms. As cirrhosis advances, liver function decreases. Cirrhosis can lead to severe complications, including hepatic encephalopathy, portal hypertension, varices, ascites, infections, and cancer [1].

Hepatic encephalopathy is a complex brain dysfunction that occurs in the context of acute liver failure, portal-systemic shunting without liver disease, and frequently in cirrhosis [2, 3, 4]. Symptoms encompass both neurological and psychiatric alterations, with severity varying from mild manifestations to coma [5, 6]. The current classification, using the terms 'covert' and 'overt' hepatic encephalopathy, was introduced in the joint 2014 guidelines of the American Association for the Study of Liver Diseases (AASLD) and the European Association for the Study of the Liver (EASL) on hepatic encephalopathy, and the classification is still recommended for use in the newest 2022 EASL update [7, 8]. People with covert hepatic encephalopathy present with minor or no clinical signs or symptoms, yet they demonstrate neuropsychiatric abnormalities diagnosed through psychometric tests [9]. Several scales exist to assess the grade of hepatic encephalopathy, with West Haven Criteria, grading from 0 to IV (subtle neurological and mental changes to coma), being the most widely used [7]. Overt hepatic encephalopathy is defined as meeting criteria of grades II or higher according to the West Haven criteria [7, 8].

Overt hepatic encephalopathy in people with cirrhosis generally occurs due to precipitating events such as variceal bleeding, hyponatraemia, dehydration, constipation, or infections [8]. All-cause mortality approaches 50% amongst people who are hospitalised with severe hepatic encephalopathy [10], while it ranges from 3% to 14% amongst people diagnosed with covert hepatic encephalopathy (including minimal hepatic encephalopathy) [11, 12]. Overt hepatic encephalopathy is classified as episodic, recurrent (more than one episode over six months), or persistent [8]. Both covert and overt hepatic encephalopathy are associated with physical symptoms and have a considerable impact on the quality of life. Early prevention can reduce the recurrence of symptoms, improve daily functioning, and enhance quality of life. Thus, effective management strategies are needed [13, 14].

Description of the intervention and how it might work

The three branched-chain amino acids (BCAAs) — leucine, isoleucine, and valine — are essential amino acids with a carbon chain [15]. Nutritional supplements containing BCAAs can be administered orally or intravenously. A typical dosage of oral BCAA supplements would be approximately 0.25 mg/kg/day [16]. The available products also differ in terms of the content, as illustrated in randomised clinical trials on BCAAs with products containing different amounts of BCAA [17, 18, 19].

The pathogenesis of hepatic encephalopathy remains poorly understood. However, ammonia plays a central role [20], and several studies report disturbed cerebral oxygen uptake and blood

flow during hepatic encephalopathy, which may be related to increased brain ammonia uptake [21]. Most interventions for hepatic encephalopathy aim to decrease the blood ammonia level [22, 23].

In both acute and chronic liver failure, skeletal muscle can take up ammonia from plasma and release glutamine [24]. Biochemically, plasma BCAAs are often depleted during hyperammonaemia as BCAAs are consumed in skeletal muscle in concert with ammonia [25, 26]. BCAAs supply the muscle tissue with carbon skeletons for replenishment of α -ketoglutarate, which may enhance amination to glutamate and, subsequently, amidation of glutamate to glutamine. In people with cirrhosis and hyperammonemia, plasma and muscle tissue concentrations of BCAAs and glutamate are diminished, and the removal of ammonia in muscle is proportional to the removal of BCAAs. Thus, skeletal muscle is believed to play a key role in ammonia detoxification and becomes an important alternative organ for ammonia detoxification.

The effects of BCAA supplements are complex, but several studies have indicated that BCAAs enhance this detoxification by supplying carbon skeletons [25, 26]. Additionally, supplementing with BCAAs might alleviate malnutrition and counteract the loss of muscle cell mass often observed in advanced liver disease, as well as reduce the protein breakdown seen in people with hepatic encephalopathy [27, 28]. BCAAs have strong anabolic properties via stimulation of protein synthesis, and the increased muscle mass may further increase extrahepatic ammonia detoxification [29]. The mechanism involves anabolic signalling pathways which increase the translation of messenger ribonucleic acid (mRNA). BCAAs, and especially leucine, have gained particular interest as a strong activator of the mammalian target of rapamycin (mTOR) pathway, which governs muscle cell growth and proliferation [30, 31]. Also, glutamine synthetase activity is high in muscle tissue, which promotes detoxification of ammonia to glutamine. BCAA supplementation also increases plasma levels of BCAAs and reduces the ratio between aromatic amino acids and BCAAs [15]. BCAAs may further enhance detoxification of ammonia in skeletal muscle by the amidation process for glutamine synthesis [32]. The addition of BCAAs reduces cerebral efflux of aromatic amino acids across the blood-brain barrier and the imbalance in the synthesis of dopamine, noradrenaline, and serotonin [33].

BCAAs may be administered orally or intravenously, each with different clinical considerations. Oral BCAAs are non-invasive and suitable for outpatient settings, although gastrointestinal absorption can potentially influence bioavailability and therapeutic effects. Intravenous BCAAs ensure complete systemic delivery but require venous access and therefore carry the recognised risks associated with intravenous (IV) therapy. These include fluid overload associated with the 500 mL of BCAA administered as well as catheter-related infections, which, while generally uncommon, are a well-documented potential complication of IV administration. The differences in bioavailability, risk profile, and tolerability between oral and intravenous administration may help explain the heterogeneity of outcomes observed across clinical studies [34].

Why it is important to do this review

Cognitive impairment associated with cirrhosis leads to greater utilisation of healthcare resources by adults than any other manifestations of liver disease [35]. Diagnostic tools vary from

basic scales to advanced psychometric tests. The efficacy of each tool and the choice of the optimal diagnostic approach depend on the specific clinical scenario. Accordingly, investigators conducting clinical trials on hepatic encephalopathy have used a variety of scales and tests in their assessments of the diagnosis and improved manifestations of hepatic encephalopathy [11, 17, 36, 37]. Determining clinically significant improvement in hepatic encephalopathy is a crucial outcome, regardless of symptom severity [4, 7, 38, 39].

Older meta-analyses of randomised trials and observational studies indicate that the evidence regarding BCAA supplements is inconclusive due to methodological concerns, small sample sizes, and short follow-up [40, 41, 42, 43, 44]. However, recent updates demonstrate that BCAAs have a favourable impact on hepatic encephalopathy, yet no effect on all-cause mortality [45, 46, 47].

The 2014 guideline from AASLD/EASL recommends treatment of any precipitation factors, non-absorbable disaccharides as initial treatment, and BCAAs as an alternative or additional agent to treat those people who do not respond to initial therapy [7, 39]. Yet, evidence that BCAAs prevent recurrence in people with a former episode of overt hepatic encephalopathy is sparse (Les 2011 [17]), and currently, in the latest guideline update from 2022, rifaximin is the first adjunct to lactulose following one additional episode of overt hepatic encephalopathy within six months of the first one [8]. Systematic reviews on interventions for hepatic encephalopathy found evidence that orally administered BCAA supplements improve hepatic encephalopathy but not mortality [48, 49].

Newly published randomised trials were expected to provide valuable new insights [50, 51]. Therefore, we performed this update of our Cochrane review, first published in 2003 [43], and previously updated in 2017 [47].

OBJECTIVES

To assess the beneficial and harmful effects of BCAAs versus any control intervention for people with cirrhosis and hepatic encephalopathy.

METHODS

In the updated review, we changed the methods and the reporting to reflect current guidelines in the *Cochrane Handbook for Systematic Reviews of Interventions* [52], the Methodological Expectations of Cochrane Intervention Reviews (MECIR) guidelines [53], and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 (PRISMA-S Checklist) [54]. Differences between the original protocol [43] and this review have been updated (see [Supplementary material 8](#)).

Criteria for considering studies for this review

Types of studies

We included randomised clinical trials irrespective of blinding, publication status, language, and the outcomes reported. We included the first period of cross-over trials. We did not expect to find cluster-randomised trials. We included trials with a subset of participants diagnosed with cirrhosis. We did not include quasi-randomised studies (i.e. trials using a method of allocation that is not truly random, such as allocation based on date of birth).

Types of participants

We included children and adults of any age, sex, race, and ethnicity, diagnosed with cirrhosis regardless of the aetiology. The included participants had hepatic encephalopathy associated with acute or chronic liver disease [7, 39]. The diagnosis could have been made based on clinical evaluations (e.g. using standardised scores), psychometric and neurophysiological scores, electroencephalography, or laboratory testing (e.g. ammonia levels). We used the diagnosis of hepatic encephalopathy as reported in the individual trials.

We included treatment-naïve participants as well as those previously treated for hepatic encephalopathy.

For trials that included only a subset of relevant participants, we attempted to obtain individual data by contacting study authors.

Types of interventions

Experimental intervention: any form of BCAA, irrespective of regimen (e.g. dose or duration).

Control intervention: no intervention, placebo, diets, non-absorbable disaccharides, antibiotics, or any other intervention with a potential effect on hepatic encephalopathy.

Outcome measures

We included trials irrespective of the reported outcomes.

Our outcomes include all core outcomes for children and adults with cirrhosis and hepatic encephalopathy. We evaluated outcomes at the longest available follow-up time point. The longest follow-up was also our primary time point for analysis.

Critical outcomes

- All-cause mortality
- Hepatic encephalopathy (treatment failure — proportion of people with cirrhosis and without improved manifestations of hepatic encephalopathy — as described by the authors of the included trials)
- Serious adverse events: any untoward medical occurrence that, at any dose, resulted in death, was life-threatening, required hospitalisation or prolongation of existing hospitalisation, resulted in persistent or significant disability or incapacity, or was a congenital anomaly/birth defect, or any medical event that might have jeopardised the person, or required intervention to prevent it [55]. In our review, we considered nausea and diarrhoea as serious adverse events [56]. We did not include mortality in the analysis of serious adverse events.

Important outcomes

- Quality of life, assessed with a validated tool such as the 36-item Short Form Health Survey (SF-36) questionnaire, EuroQoL visual analogue scale (EQ-VAS), and the Chronic Liver Disease Questionnaire (CLDQ). When a trial reported multiple health-related quality of life measures, we planned to use these in the following order: SF-36, EQ-VAS, CLDQ.
- Markers of nutritional status, including serum albumin and nitrogen balance (the nitrogen input minus the nitrogen output)

Search methods for identification of studies

To minimise bias in our search results, we followed the guidance in Chapter 4 of the *Cochrane Handbook for Systematic Reviews of Interventions* and PRISMA-S (PRISMA-S Checklist) to plan and describe the search process for the review [54, 57]. The Cochrane Hepato-Biliary Group Information Specialist developed the search strategies and performed the electronic searches. We imposed no restrictions on the publication language, date, or status.

Electronic searches

Trials were identified from electronic searches in the Cochrane Hepato-Biliary Group Controlled Trials Register via the Cochrane Central Register of Studies Web, the Cochrane Central Register of Controlled Trials (CENTRAL) in the Cochrane Library, MEDLINE ALL Ovid, Embase Ovid, LILACS (Latin American and Caribbean Health Science Information database; VHL Regional Portal), Science Citation Index Expanded, and Conference Proceedings Citation Index – Science. The latter two were searched simultaneously through Web of Science. Searches were run from inception and de-duplicated from the search of the last published review.

We used the Cochrane Highly Sensitive Search Strategies for identifying randomised trials in MEDLINE and Embase, modified to include relevant search terms for identifying retraction statements and errata, as recommended in Chapter 4 of the *Cochrane Handbook for Systematic Reviews of Interventions* [57, 58].

All search strategies are presented in [Supplementary material 1](#). This update includes searches up to 12 December 2024.

Searching other resources

We searched the online World Health Organization (WHO) International Clinical Trial Registry Platform (www.who.int/ictpr) for ongoing or unpublished trials and study information. We contacted experts in the field for information about any additional published or unpublished trials.

We searched for relevant grey literature sources such as reports, dissertations, theses, and conference abstracts in Google Scholar (using the terms 'branched chain amino acids' AND cirrhosis* AND random*).

We manually checked citations and reference lists of the included trials, and any relevant systematic reviews identified. We also manually searched for studies in specialist journals (*Hepatology* and *Journal of Hepatology*) and conference proceedings (AASLD and EASL) from 2017 (the year of our last published review) and onwards.

We searched for any relevant retraction statements (through the Retraction Watch Database (retractionwatch.com/retraction-watch-database-user-guide/)) and errata for information, as errata can reveal important limitations or even fatal flaws in studies [57].

The most recent searches in other resources were carried out on 03 January 2025.

Data collection and analysis

We followed the instructions in the *Cochrane Handbook for Systematic Reviews of Interventions* for data collection and analysis [52].

At least two review authors (LA, ND, GD, and LG) independently extracted data from the published trial reports. We wrote to authors of included trials for additional information, including data that would allow recalculation of outcomes when necessary. For trials described in more than one record, we checked the multiple publications for discrepant information. We used the publication with the most complete information (the largest number of participants and events, at the longest duration of follow-up) and contacted the authors if relevant information was not identical.

We included the trials irrespective of the reported outcomes.

Selection of studies

We listed trials identified through electronic and manual searches, selected trials using the criteria described above, and listed excluded studies with the reason for exclusion (see [Supplementary material 3](#)).

We recorded the selection process in sufficient detail to complete a PRISMA flow diagram, including searches of databases, registers, and other sources [59, 60]. We searched for any relevant retraction statements and errata for the included trials to avoid their potential impact on the overall estimates of our review [57].

Data extraction and management

LA and ND piloted the data extraction forms, and extracted the following data from each study:

- participant characteristics: inclusion criteria, mean age, proportion of men, and type of underlying liver disease;
- intervention characteristics: type, dose, and duration of interventions;
- trial characteristics: setting (hospital or outpatient), number of clinical sites, country of origin, trial participant inclusion period, and funding;
- critical and important outcomes data.

Risk of bias assessment in included studies

Two review authors (LA and ND) independently assessed the risk of bias in the trials using the RoB 2 framework [61, 62, 63]. We resolved disagreements through discussion.

We assessed the effect of assignment to the intervention (the intention-to-treat (ITT) principle). ITT includes all randomised participants, irrespective of the interventions that participants received.

We assessed the risk of bias of the following outcome results at their maximum follow-up time points: all-cause mortality, hepatic encephalopathy, serious adverse events, quality of life, and markers of nutritional status.

We used the RoB 2 Excel tool to conduct the assessment (www.riskofbias.info) [64]. We evaluated the risk of bias in the outcome results according to the five RoB 2 domains:

- bias arising from the randomisation process;
- bias due to deviations from intended interventions;
- bias due to missing outcome data;
- bias in measurement of an outcome;
- bias in selection of the reported result.

Each domain is assessed as 'high risk of bias,' 'some concerns,' or 'low risk of bias,' based on responses to signalling questions and algorithms within the RoB 2 Excel tool. We used the tool algorithm to obtain an overall risk of bias assessment for each outcome. We quoted text to support our judgments, and if we disagreed with a judgment recommended by the algorithm, we included an explicit statement of the disagreement.

For the trials with a cross-over design, we only used data from the first period. Therefore, we did not utilise the tool extension to address supplementary factors related to data from such trials [65].

Our risk of bias assessment fed into the first domain of the certainty of evidence in the GRADE summary of findings table.

Measures of treatment effect

We followed the guidance outlined in Chapters 9 and 10 of the *Cochrane Handbook for Systematic Reviews of Interventions* [66, 67]. We used risk ratios (RR) for dichotomous outcomes, standardised mean differences (SMDs) for continuous outcomes in trials using different scales, and mean differences (MDs) for continuous outcomes when trials used the same scales, all with 95% confidence intervals (CI).

The interpretation of SMDs was as follows: SMD less than 0.40 for small intervention effects; SMD between 0.40 and 0.70 for moderate intervention effects; and SMD greater than 0.70 for large intervention effects [68]. We intended to describe in a narrative format skewed data reported as medians and interquartile ranges, as described in Section 10.5.3 of the *Cochrane Handbook for Systematic Reviews of Interventions* [66].

We followed the guidance in Section 19.5.1 and 19.5.2 of the *Cochrane Handbook* when we reported on adverse events in the summary of findings tables [69].

Though planned, we did not use Peto's odds ratio to deal with rare events in binary outcomes (e.g. adverse events) because the assumptions for its use were not met in our dataset.

If mortality was reported as time-to-event, we planned to use hazard ratios (HR) [66].

We analysed the data at the end of follow-up for all outcomes.

Unit of analysis issues

We followed the guidance in the *Cochrane Handbook* to avoid unit of analysis errors [70].

The unit of analysis was the individual participant randomised to the intervention groups. For trials with a parallel-group design with multiple intervention groups, we combined all relevant experimental intervention groups into a single group, as recommended in Chapter 6 of the *Cochrane Handbook* [70], and compared the experimental intervention group to the control group. For trials with more than one control group (e.g. trials with two different control diets), we combined data from the two control groups.

For randomised trials with a cross-over design, we included only data from the first period of the trial (i.e. before the cross-over) to avoid carry-over effects [71].

We planned to include trials with mixed populations if they reported their data separately, or if we could obtain separate data from the trial authors [72].

Dealing with missing data

We gathered data to allow intention-to-treat analyses; that is, our analyses included data on all participants randomised, irrespective of compliance, protocol violations, or follow-up.

We contacted the corresponding authors of trials to verify key study characteristics and obtain missing numerical outcome data.

Reporting bias assessment

For meta-analyses with at least 10 trials, we assessed reporting biases and other dissemination biases based on analyses of small-study effects [73], using the Harbord modified test for dichotomous outcomes and Egger's tests for continuous outcomes [74, 75]. The Harbord test regresses $Z/\sqrt{V}$ against $\sqrt{V}$, where Z is the efficient score and V is Fisher's information (the variance of Z under the null hypothesis). Egger's test performs a linear regression of the intervention effect estimates on their standard errors, weighing by $1/(\text{variance of the intervention effect estimate})$. We also attempted to obtain trial protocols to compare prespecified and reported outcome measures.

Synthesis methods

We performed our analyses using Review Manager (RevMan) and STATA [76, 77]. We did not limit our primary analysis to trials at an overall low risk of bias.

We performed analyses according to the recommendations in Chapter 10 of the *Cochrane Handbook* [66].

We used a random-effects model as we expected clinical diversity. We used the Restricted Maximum Likelihood (REML) estimator to estimate between-trial variance. We used the Hartung-Knapp-Sidik-Jonkman (HKSJ) method to calculate a CI for the meta-analysis effect estimate when there were at least three trials and the estimate of heterogeneity was greater than zero. In other scenarios (i.e. in pooled analyses of two trials or where the estimate of heterogeneity equalled zero), we used the Wald-type method.

We expressed heterogeneity as I^2 values using these thresholds: 0% to 40% (unimportant), 40% to 60% (moderate), 60% to 80% (substantial), and greater than 80% (considerable).

Investigation of heterogeneity and subgroup analysis

We planned the following subgroup analyses of the [Critical outcomes](#) to identify potential effect modifiers, irrespective of the presence or absence of heterogeneity:

- mode of administration (oral or intravenous);
- type of hepatic encephalopathy (overt or covert);
- type of control group (e.g. no intervention, placebo, diets, non-absorbable disaccharides, or antibiotics);
- type of data (recalculated and trial reports data).

We assessed subgroup effects using P values from tests for subgroup differences, and considered a P value of less than 0.1 as indicative of a significant subgroup effect.

Equity-related assessment

We acknowledge the importance of health equity. However, we did not conduct formal equity-related assessments in this review due to limited reporting of PROGRESS-Plus variables (e.g. occupation, socioeconomic status, ethnicity, and education) in the included trials.

Sensitivity analysis

To evaluate the robustness of the overall results, we performed the following sensitivity analyses on the critical outcomes:

- repeating the analyses with the fixed-effect model;
- including only trial results judged to be at an overall low risk of bias;
- including only trials published as full-text, peer-reviewed articles.

Exploratory (post hoc) analyses

Based on peer-review comments from a previous update [45], we performed sensitivity analyses in which we excluded:

- trials with a no-intervention, placebo, or diet control group (i.e. only included trials with a lactulose or neomycin control group);
- trials with a lactulose or neomycin control group;
- trials that included people with or without cirrhosis.

Certainty of the evidence assessment

We followed the GRADE approach to prepare a summary of findings table (Summary of findings 1), using GRADEpro software [78, 79], presenting information on the results of our critical and important outcomes: all-cause mortality, hepatic encephalopathy, serious adverse events, quality of life, and markers of nutritional status. We combined all control groups (no intervention, placebo, diets, and other medical treatments for hepatic encephalopathy).

Two review authors (LA and ND) independently assessed the certainty of the evidence. LA and ND resolved any disagreements through discussion or, whenever necessary, by consulting a third review author (LLG).

We used the five GRADE factors – risk of bias (study limitations), inconsistency, imprecision (using the minimally contextualised approach) [80; 81], indirectness, and publication bias – to assess the certainty of the body of evidence in the meta-analysed outcomes. We downgraded the evidence by one level for serious concerns and two levels for very serious concerns.

We judged the risk of bias per outcome, and we used the overall judgement for the outcome result. Low risk meant no study limitations, some concerns indicated possible or serious limitations (downgraded by one level), and high risk reflected serious or very serious limitations (downgraded by two levels) [80].

To assess imprecision, we followed the latest guidance using a minimally contextualised approach and focused on absolute effect confidence intervals (CIs) and thresholds [81]. We considered 0.25 as a trivial threshold of interest. When (a) the CI crossed one threshold of interest, we downgraded one level for imprecision; (b) when the CI crossed the thresholds of important benefit and important harm, we downgraded two levels for imprecision; (c) when the CI was very wide and both boundaries of the CI suggested

very different inferences, we planned to downgrade three levels; and (d) when the CI did not cross the threshold and the relative effect was large, we implemented the optimal information size (OIS) approach, rating down more than one level for imprecision when the sample size of the meta-analysis was far less than the OIS [80].

GRADE levels of evidence are defined as 'high', 'moderate', 'low', or 'very low', and interpreted as follows.

- **High certainty:** we are very confident that the true effect lies close to that of the estimate of the effect.
- **Moderate certainty:** we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
- **Low certainty:** our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
- **Very low certainty:** we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

We elaborated all decisions for downgrading the certainty of the evidence in the footnotes of the summary of findings table. We included additional comments in the comment field of the serious adverse events analysed.

Consumer involvement

Consumers were not involved in this review.

RESULTS

Description of studies

Table 1 provides a summary of included studies and syntheses. Key study characteristics, which are important for interpreting the syntheses, and full study details are available in Supplementary material 2.

We added two new randomised trials with 108 participants to this review (Vidot 2019 [51, 82, 83, 84]; Mehtani 2023 [50, 85, 86, 87]), making a total of 18 randomised clinical trials, published between 1984 and 2023, in qualitative and quantitative analyses (Calvey 1985 [88, 89, 90]; Cerra 1985 [91, 92, 93, 94]; Egberts 1985 [36, 95, 96, 97, 98]; Fiaccadori 1985 [99, 100, 101, 102, 103]; Hayashi 1991 [104]; Horst 1984 [105, 106]; Hwang 1988 [107]; Les 2011; Marchesini 1990 [37, 108, 109]; Marchesini 2003 [11]; Mehtani 2023; Michel 1985 [110, 111, 112, 113]; Muto 2005 [12, 114, 115]; Plauth 1993 [116, 117]; Rossi-Fanelli 1986 [118, 119, 120, 121]; Strauss 1986 [122, 123]; Vidot 2019; Vilstrup 1990 [124, 125, 126]). In total, the trials included 934 participants with hepatic encephalopathy. All participants had cirrhosis, except for one trial including participants with acute hepatitis (Calvey 1985).

Results of the search

The previous version of the review included 16 trials (40 references). Updated searches were performed on 12 December 2024 (Figure 1). We identified a total of 4900 potentially relevant records in the electronic searches and 20 additional records in the manual searches (Figure 1). After removing 1181 duplicates, we screened 3739 records. Based on title, abstract, or both, we excluded 3666 records. Of the remaining 73 full-text records, we excluded 21

studies (22 records), leaving 18 randomised clinical trials (48 records) that fulfilled our inclusion criteria. We found one trial published in abstract form (Hayashi 1991), and 17 trials described in full-text articles (Fiaccadori 1985; Horst 1984; Calvey 1985; Cerra 1985; Egberts 1985; Michel 1985; Rossi-Fanelli 1986; Strauss 1986; Hwang 1988; Marchesini 1990; Vilstrup 1990; Plauth 1993; Marchesini 2003; Muto 2005; Les 2011; Vidot 2019; Mehtani 2023). We were able to obtain unpublished information about the trial

design and data that allowed re-calculation of outcomes based on the individual participant data for four trials (Marchesini 1990; Marchesini 2003; Muto 2005; Les 2011; see [Supplementary material 2](#)). Three authors of the present review (IL, GM, and HV) were investigators in four included studies (Marchesini 1990; Vilstrup 1990; Marchesini 2003; Les 2011). They were not involved with the selection, data extraction, risk of bias, or GRADE assessment of their trials.

Figure 1. PRISMA flow diagram [59, 60]. Date of last search: 12 December 2024

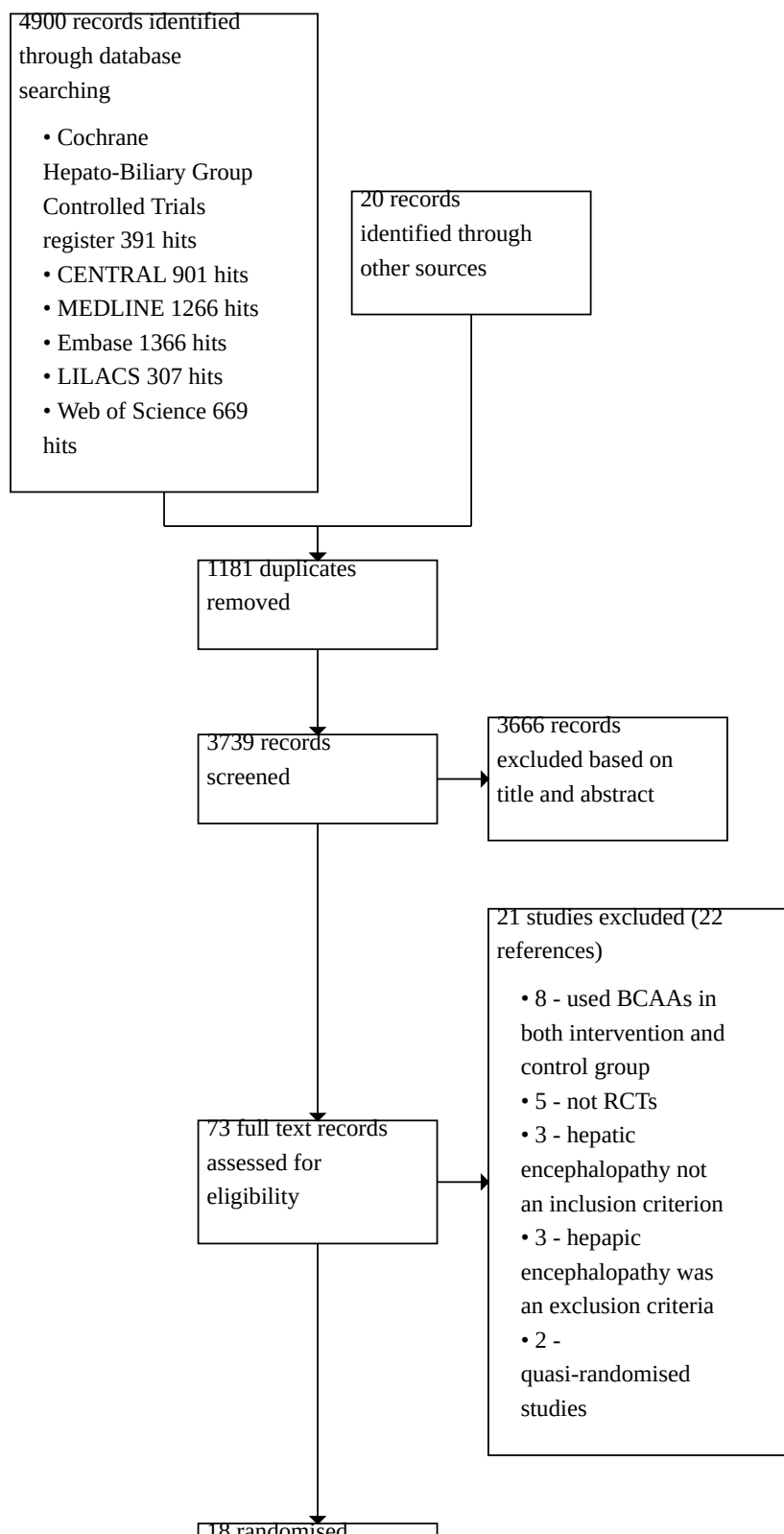
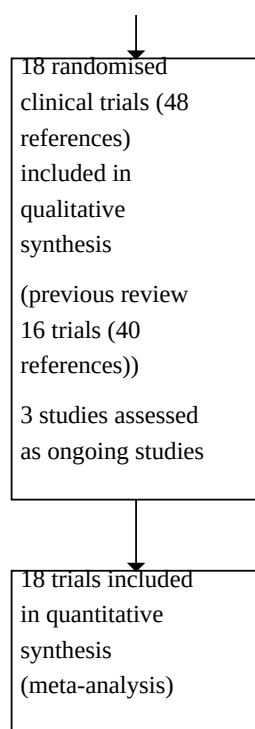


Figure 1. (Continued)



We identified three ongoing trials (CTRI/2019/08/020552 [127]; NCT05700695 [128]; NCT06538077 [129]; see [Supplementary material 4](#) for details).

Included studies

Trial design

Sixteen trials had a parallel-group randomised clinical trial design (Calvey 1985; Cerra 1985; Fiaccadori 1985; Hayashi 1991; Horst 1984; Hwang 1988; Les 2011; Marchesini 1990; Marchesini 2003; Mehtani 2023; Michel 1985; Muto 2005; Rossi-Fanelli 1986; Strauss 1986; Vidot 2019; Vilstrup 1990), and two trials used a cross-over randomised clinical trial design (Egberts 1985; Plauth 1993). The follow-up periods in the trials were from four days to two years (104 weeks).

Participants

The trials included 934 adults with hepatic encephalopathy. None used BCAA deficiency as an inclusion or exclusion criterion. Most participants had cirrhosis (minimum 97%). One trial, Calvey 1985, included 64 people with acute liver disease with or without cirrhosis. Of these 64 participants, 30 had cirrhosis, and we included only 26 people with overt hepatic encephalopathy at baseline in our analyses. The proportion of men ranged from 47% to 90%, the proportion of people with alcoholic liver disease ranged from 8% to 100%, and the proportion of people with viral hepatitis ranged from 0% to 81%. The mean age ranged from 47 to 64 years. In three trials, 6% to 74% of the trial participants had hepatic encephalopathy at baseline (in our analysis, we included data only from those with hepatic encephalopathy) (Marchesini 2003; Muto 2005; Les 2011). The type of hepatic encephalopathy was minimal in five

trials (Egberts 1985; Plauth 1993; Marchesini 2003; Les 2011; Vidot 2019), and overt in the remaining 13 trials (Calvey 1985; Cerra 1985; Fiaccadori 1985; Hayashi 1991; Horst 1984; Hwang 1988; Marchesini 1990; Mehtani 2023; Michel 1985; Muto 2005; Rossi-Fanelli 1986; Strauss 1986; Vilstrup 1990). We found no trials in children. Ten trials were conducted in Europe, four in Asia, two in the USA, and one each in Brazil and Australia.

Interventions

In one trial, most participants received oral BCAAs, and those who deteriorated received intravenous BCAAs (Calvey 1985). Ten trials assessed oral BCAAs (Calvey 1985; Egberts 1985; Hayashi 1991; Horst 1984; Les 2011; Marchesini 1990; Marchesini 2003; Muto 2005; Plauth 1993; Vidot 2019), and eight trials, intravenous BCAAs (Cerra 1985; Fiaccadori 1985; Hwang 1988; Mehtani 2023; Michel 1985; Rossi-Fanelli 1986; Strauss 1986; Vilstrup 1990). Trials studied intravenous BCAA-treated people in the hospital and the remainder on an outpatient basis. The dose of BCAAs ranged from 6 g/day to 57 g/day, and the treatment duration was from one to 104 weeks (median four weeks).

Controls

Ten trials used diets as the control intervention (Calvey 1985; Egberts 1985; Hayashi 1991; Horst 1984; Les 2011; Marchesini 1990; Marchesini 2003; Michel 1985; Muto 2005; Vilstrup 1990). Three trials used placebo or no intervention as the control (Hwang 1988; Plauth 1993; Vidot 2019). Five trials used other medical treatments for hepatic encephalopathy as the control intervention(s), as follows: testing intravenous BCAAs with or without lactulose versus lactulose alone (Fiaccadori 1985); intravenous BCAAs versus

lactulose (Rossi-Fanelli 1986); intravenous BCAAs plus lactulose versus lactulose alone (Mehtani 2023); or intravenous BCAAs versus neomycin (Cerra 1985; Strauss 1986).

Co-interventions

One trial administered lactulose equally to the intervention group and control group (Mehtani 2023).

Outcomes

For trials that fulfilled the inclusion criteria but did not report on one or more of the outcomes of interest, we contacted the study authors to request additional information (see Notes of [Supplementary material 2](#)). One trial did not report all-cause mortality (Hayashi 1991). The remaining trials reported all-cause mortality at the end of treatment. All trials reported the effect of BCAAs on hepatic encephalopathy. Thirteen trials assessed the effect of BCAAs on manifestations of overt hepatic encephalopathy (Calvey 1985; Cerra 1985; Fiaccadori 1985; Hayashi 1991; Horst 1984; Hwang 1988; Marchesini 1990; Mehtani 2023; Michel 1985; Muto 2005; Rossi-Fanelli 1986; Strauss 1986; Vilstrup 1990), and five trials of minimal hepatic encephalopathy (Egberts 1985; Les 2011; Marchesini 2003; Plauth 1993; Vidot 2019). Nausea and diarrhoea were reported in six trials (Hayashi 1991; Horst 1984; Marchesini 2003; Mehtani 2023; Muto 2005; Plauth 1993), albumin in three trials (Les 2011; Marchesini 1990; Marchesini 2003), and nitrogen balance in three trials (Cerra 1985; Egberts 1985; Marchesini 1990).

Dropouts or withdrawals

Four trials did not account for participants with missing outcome data (Fiaccadori 1985; Hayashi 1991; Rossi-Fanelli 1986; Vilstrup 1990). The remaining trials reported outcome data for all randomised participants (Calvey 1985; Cerra 1985; Egberts 1985; Horst 1984; Hwang 1988; Les 2011; Marchesini 1990; Marchesini 2003; Mehtani 2023; Michel 1985; Muto 2005; Plauth 1993; Strauss 1986; Vidot 2019).

Intention-to-treat

Seven trials specifically pointed out the use of intention-to-treat analyses (Hwang 1988; Marchesini 2003; Mehtani 2023; Michel 1985; Muto 2005; Vidot 2019; Vilstrup 1990). We contacted investigators if we were unsure whether intention-to-treat analyses were used. We contacted 11 trial investigators for data on design, number

of participants, or data (Egberts 1985; Fiaccadori 1985; Les 2011; Marchesini 1990; Marchesini 2003; Mehtani 2023; Muto 2005; Rossi-Fanelli 1986; Strauss 1986; Vidot 2019; Vilstrup 1990), and received responses from all.

Funding

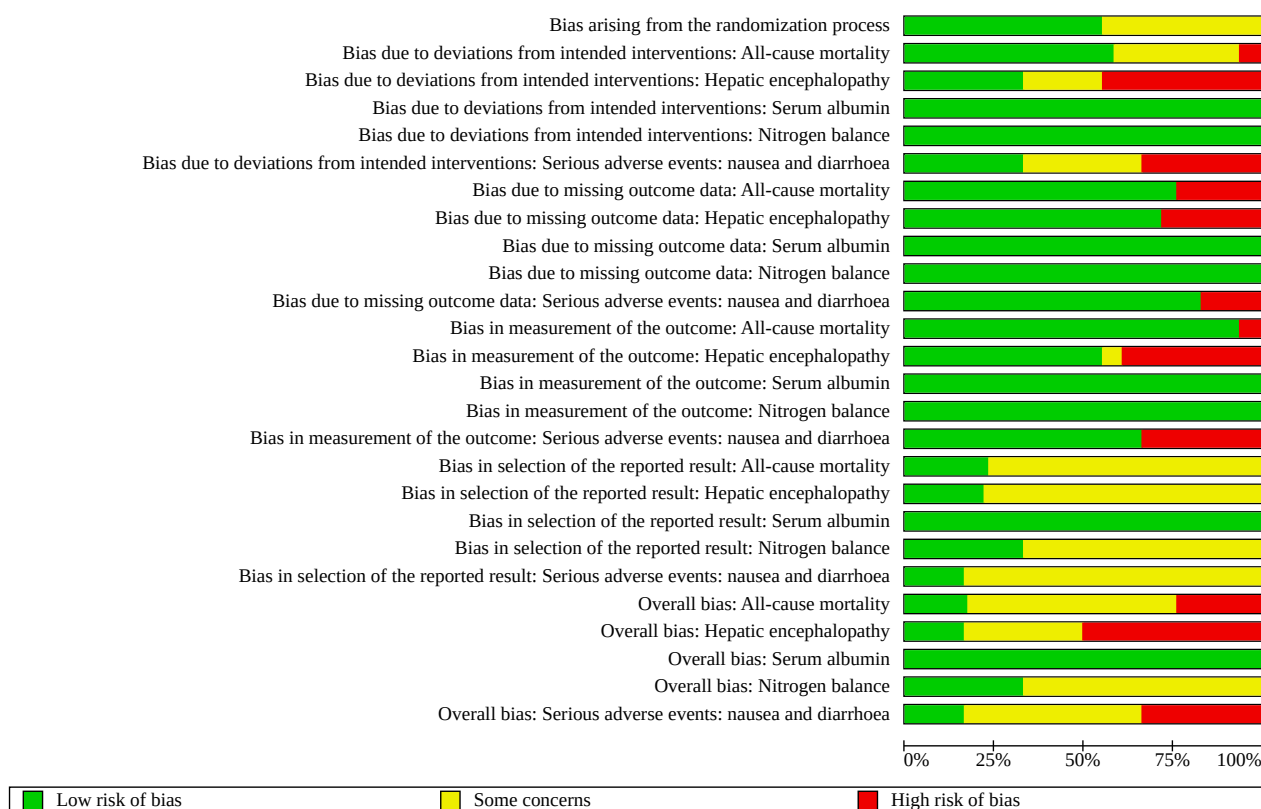
Three trials received support from for-profit organisations in the form of interventions (Muto 2005; Plauth 1993; Vilstrup 1990). None of the remaining 15 trials received support from for-profit organisations (Calvey 1985; Cerra 1985; Egberts 1985; Fiaccadori 1985; Hayashi 1991; Horst 1984; Hwang 1988; Les 2011; Marchesini 1990; Marchesini 2003; Mehtani 2023; Michel 1985; Rossi-Fanelli 1986; Strauss 1986; Vidot 2019).

Excluded studies

We excluded 21 trials (22 references), with reasons listed in [Supplementary material 3](#). Briefly, eight trials used BCAAs in both intervention and control groups (Christie 1985 [130]; Espina 2021 [131]; Hayashi 2007 [132]; Malaguarnera 2009 [133]; O'Keefe 1987 [134]; Romeiro 2018 [135]; Tejavath 2021 [136]; Walker 1982 [137]), five were not randomised clinical trials (Okada 2020 [138]; Okada 2022 [139]; Park 2017 [140]; Park 2020 [141, 142]; Yoshida 1989 [143]), three did not have hepatic encephalopathy as an inclusion criterion (Kanematsu 1988 [144]; Kobayashi 2008 [145]; Nojiri 2017 [146]), three had hepatic encephalopathy as an exclusion criterion (Kawamura 2009 [147]; Nakaya 2007 [148]; Poon 2004 [149]), and two were quasi-randomised studies (Eriksson 1982 [150]; Wahren 1983 [151]).

Risk of bias in included studies

We used the RoB 2 tool to assess the risk of bias [61, 62]. Results are graphically presented for each outcome alongside the forest plots in the analyses (Analysis 1.1; Analysis 1.3; Analysis 1.4; Analysis 1.5; Analysis 1.2). The assessment of bias included information retrieved from the published trial reports as well as information from the primary investigators ([Supplementary material 2](#); [Figure 2](#)). We have included risk of bias assessments and support for judgements across all RoB 2 domains for each outcome specified for risk of bias assessment in [Supplementary material 5](#). For the two cross-over trials (Egberts 1985; Plauth 1993), we only used data from the first period. RoB 2 consensus responses are available by contacting the review authors.

Figure 2. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included trials.

Seventeen trials reported all-cause mortality. We judged three trials to be at low risk of bias (Les 2011; Marchesini 1990; Marchesini 2003), 10 trials to raise some concerns as no protocols were available, there was insufficient information regarding the randomisation process, and no blinding (not considered to influence mortality) (Calvey 1985; Cerra 1985; Egberts 1985; Horst 1984; Hwang 1988; Mehtani 2023; Michel 1985; Muto 2005; Plauth 1993; Strauss 1986), and four trials to be at high risk of bias due to no information on randomisation, allocation concealment, and blinding of participants, personnel and outcome assessment, and not all participants were accounted for (Fiaccadori 1985; Rossi-Fanelli 1986; Vidot 2019; Vilstrup 1990).

Eighteen trials reported hepatic encephalopathy. We judged three trial results to be at low risk of bias (Les 2011; Marchesini 1990; Marchesini 2003), six to raise some concerns as no protocols were available and there was insufficient information regarding the randomisation process (Calvey 1985; Cerra 1985; Egberts 1985; Horst 1984; Mehtani 2023; Plauth 1993), and nine trials to be at high at risk of bias because not all participants were accounted for, there was no information on the randomisation process, and blinding of participants, personnel, and/or outcome assessments was lacking (Fiaccadori 1985; Hayashi 1991; Hwang 1988; Michel 1985; Muto 2005; Rossi-Fanelli 1986; Strauss 1986; Vidot 2019; Vilstrup 1990).

Six trials reported on serious adverse events, including nausea and diarrhoea. We judged one trial result to be at low risk of bias (Marchesini 2003), three trials raised some concerns because no protocols were available and there was insufficient information

regarding the randomisation process (Horst 1984; Mehtani 2023; Plauth 1993), and two trials to be at high risk of bias as not all participants were accounted for, there was no information on the randomisation process, and blinding of participants, personnel, and/or outcome assessments was lacking (Hayashi 1991; Muto 2005).

Three trials reported serum albumin levels. We judged all trial results to be at low risk of bias (Les 2011; Marchesini 1990; Marchesini 2003).

Three trials reported nitrogen. We judged one trial result to be at low risk of bias (Marchesini 1990), and two raised some concerns because no protocols were available (Cerra 1985; Egberts 1985).

Synthesis of results

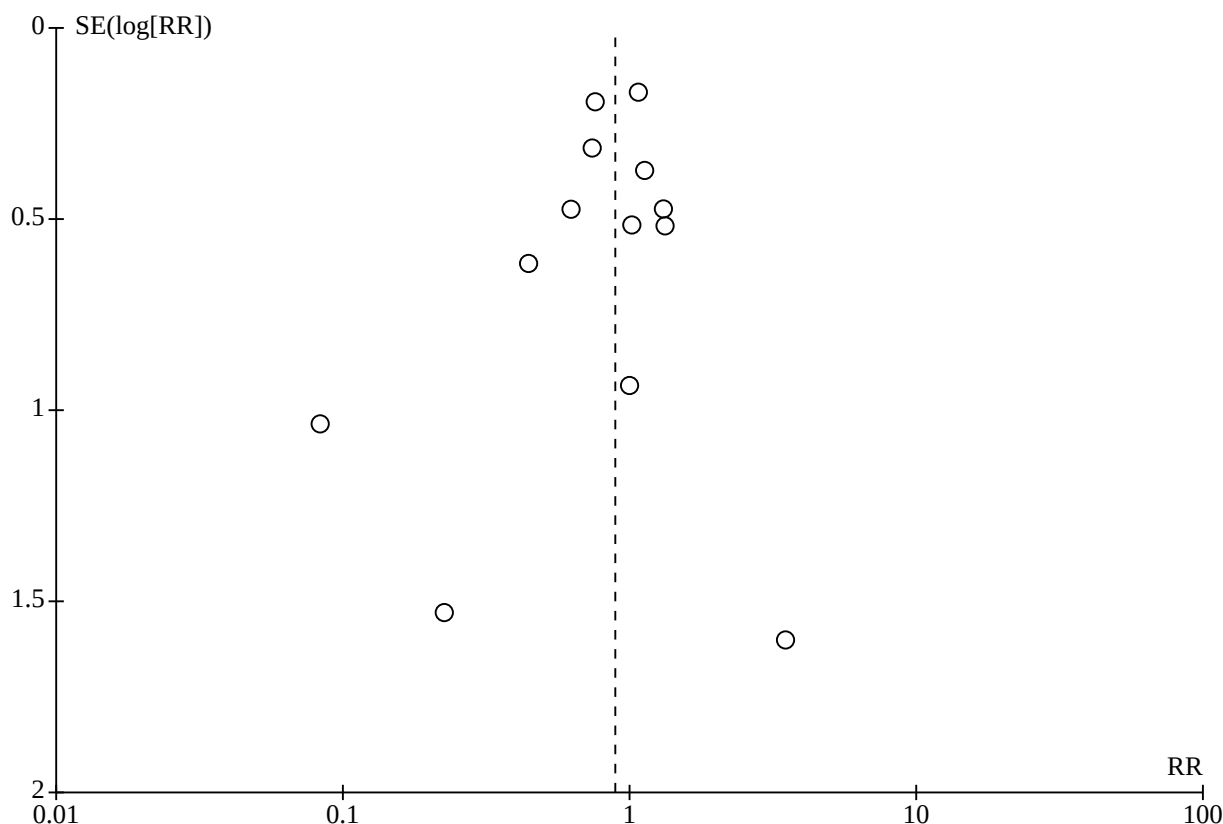
A complete record of all comparisons and analyses is available for this review in [Supplementary material 6](#) and a full data package is available in [Supplementary material 7](#).

All-cause mortality

Compared with the control interventions, BCAAs may have little to no effect on all-cause mortality, but the evidence is very uncertain (RR 0.89, 95% CI 0.71 to 1.12; $I^2 = 1\%$; 17 studies, 867 participants; very low-certainty evidence; Analysis 1.1 in [Supplementary material 6](#)). Heterogeneity was unimportant ($\tau^2 = 0.01$; $\chi^2 = 12.61$, $df = 12$ ($P = 0.40$); $I^2 = 1\%$). No evidence of small-

study effects was identified in a funnel plot (Figure 3). The range of mortality follow-up was four days to two years.

Figure 3. Funnel plot assessing publication bias for trials reporting all-cause mortality outcomes

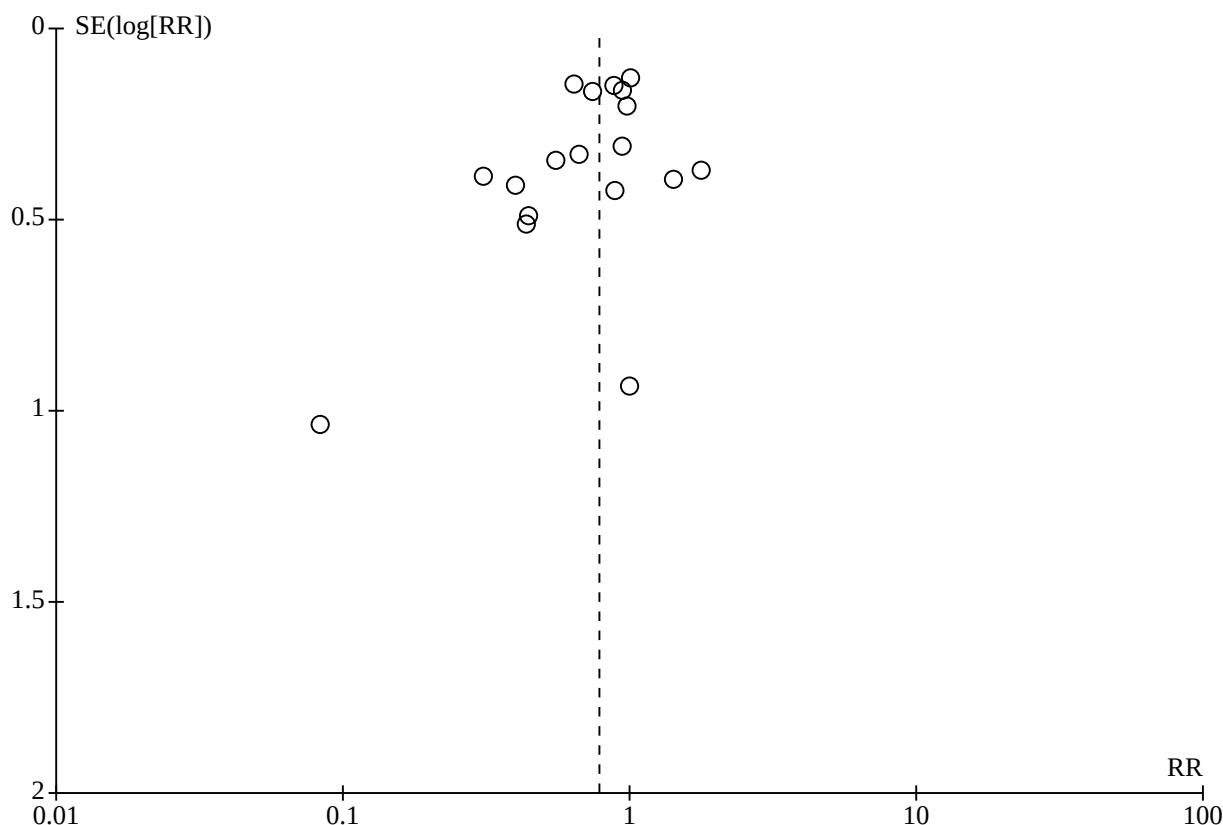


We downgraded the certainty of evidence to very low for risk of bias (two levels because most of the included trials were judged to be at high risk of bias or with some concerns), and for imprecision in the overall assessment (two levels because the CI is wide, crossing the null effect, including important benefit and unimportant harm).

Hepatic encephalopathy

All trials reported data on hepatic encephalopathy (Analysis 1.2 in [Supplementary material 6](#)). The evidence suggests that

BCAAs, compared with controls, reduce hepatic encephalopathy slightly (RR 0.79, 95% CI 0.64 to 0.96; $I^2 = 40\%$; 18 studies, 934 participants; low-certainty evidence), but the evidence is uncertain. The heterogeneity was moderate ($\text{Tau}^2 = 0.04$; $\text{Chi}^2 = 33.28$, $\text{df} = 17$ ($P = 0.01$); $I^2 = 40\%$). No evidence of small-study effects was identified in a funnel plot (Figure 4).

Figure 4. Funnel plot assessing publication bias for trials reporting hepatic encephalopathy outcomes

We downgraded the certainty of evidence to low for risk of bias (two levels because most of the included trials were judged to be at high risk of bias or with some concerns).

Serious adverse events

One trial reported serious adverse events (apart from all-cause mortality) (Mehtani 2023). This trial included participants with acute-on-chronic liver failure, and considered acute kidney injury requiring dialysis (BCAAs $n = 2/35$; control $n = 4/35$) and need for ventilation (BCAAs $n = 11/35$; control $n = 16/35$) as serious adverse events.

Common adverse events, such as nausea and diarrhoea, which often may be considered as both serious and non-serious adverse events, occurred in 58 of 477 (12%) participants in the BCAA group and 16 of 538 (3%) participants randomised to diets, i.e. the control intervention (Analysis 1.3 in [Supplementary material 6](#)). The evidence is very uncertain about the effect of BCAAs on nausea and diarrhoea (RR 2.05, 95% CI 0.40 to 10.58; $I^2 = 50\%$; 6 studies, 1015 participants; very low-certainty evidence). Heterogeneity was considerable ($\text{Tau}^2 = 0.49$; $\text{Chi}^2 = 9.76$, $\text{df} = 4$ ($P = 0.04$); $I^2 = 50\%$).

We downgraded the certainty of evidence to very low for risk of bias (two levels because most of the included trials were judged to be at high risk of bias or with some concerns), and for imprecision in the overall assessment (one level because the CI is wide, crossing the null effect, including important benefit and important harm).

Quality of life

Three trials examining oral BCAA supplements assessed quality of life scores at baseline and end of treatment, based on the 36-item Short Form Health Survey (SF-36) (Marchesini 2003; Muto 2005; Les 2011). Due to the methods used to register the scores, we were unable to combine the results in meta-analyses. All participants in the three trials had cirrhosis, but some did not have hepatic encephalopathy at baseline. When the analyses were limited to people with hepatic encephalopathy, none of the trials found beneficial or harmful effects of BCAAs on the global SF-36 score or any of the subscales.

Nutritional outcomes

BCAAs may have little to no effect on post-treatment albumin, but the evidence is very uncertain (MD 0.60, 95% CI -0.90 to 2.09; $I^2 = 0\%$; 3 studies, 176 participants; very low-certainty evidence; Analysis 1.4 in [Supplementary material 6](#)) (Les 2011; Marchesini 1990; Marchesini 2003). Heterogeneity was unimportant ($\text{Tau}^2 = 0.00$; $\text{Chi}^2 = 1.02$, $\text{df} = 2$ ($P = 0.60$); $I^2 = 0\%$).

We downgraded the certainty of evidence to very low for indirectness (one level because this is an indirect outcome used as a surrogate estimate of nutritional status), imprecision in the overall assessment (two levels because the CI crosses the null effect, and sample size does not meet the optimal information size (OIS) of 400 participants), and for publication bias (because of the few trials, we were unable to determine the risk of publication bias, but it may be strongly suspected).

The evidence is very uncertain about the effect of BCAAs on nitrogen balance, based on evidence from one trial with a neomycin control group and two trials with isonitrogenous controls (SMD 0.82, 95% CI -1.01 to 2.64; $I^2 = 69\%$; 3 studies, 108 participants; very low-certainty evidence; Analysis 1.5 in [Supplementary material 6](#)) (Cerra 1985; Egberts 1985; Marchesini 1990). Heterogeneity was considerable ($\text{Tau}^2 = 0.33$; $\text{Chi}^2 = 5.73$, $\text{df} = 2$ ($P = 0.06$); $I^2 = 69\%$).

We downgraded the certainty of evidence to very low for risk of bias (one level because we had some concerns about bias in the selection of reported results in two of the three trials), indirectness (one level because of the indirect outcome used as a surrogate estimate of nutritional status), imprecision in the overall assessment (two levels because the CI crosses the null effect, and sample size does not meet the OIS of 400 participants), and publication bias (because of the few trials, we were unable to determine the risk of publication bias, but it may be strongly suspected).

One further trial measured nitrogen balance as an outcome (Calvey 1985). This trial demonstrated no difference in nitrogen balance between the BCAA and control groups. We could not include the results in the meta-analysis on nitrogen balance due to some participants also having non-cirrhotic liver disease.

Subgroup analysis and investigation of heterogeneity

All-cause mortality

Subgroup analyses of all-cause mortality found no subgroup differences between oral and intravenous BCAAs (Analysis 1.6 in [Supplementary material 6](#); test for subgroup differences: $\text{Chi}^2 = 0.00$, $\text{df} = 1$ ($P = 0.96$), $I^2 = 0\%$), overt and minimal hepatic encephalopathy (Analysis 1.7 in [Supplementary material 6](#); test for subgroup differences: $\text{Chi}^2 = 0.60$, $\text{df} = 1$ ($P = 0.44$), $I^2 = 0\%$), different types of control interventions (Analysis 1.8 in [Supplementary material 6](#); test for subgroup differences: $\text{Chi}^2 = 5.63$, $\text{df} = 4$ ($P = 0.23$), $I^2 = 28.9\%$), or the type of data (Analysis 1.9 in [Supplementary material 6](#); test for subgroup differences: $\text{Chi}^2 = 0.02$, $\text{df} = 1$ ($P = 0.89$), $I^2 = 0\%$).

We avoided conducting a subgroup analysis to evaluate the effect of publication status since only trials published as full-text articles reported all-cause mortality.

Hepatic encephalopathy

Subgroup analysis found no subgroup differences between oral BCAAs and intravenous BCAAs (Analysis 1.10 in [Supplementary material 6](#); test for subgroup differences: $\text{Chi}^2 = 1.19$, $\text{df} = 1$ ($P = 0.27$), $I^2 = 16.2\%$), trials stratified by type of hepatic encephalopathy (Analysis 1.11 in [Supplementary material 6](#); test for subgroup differences: $\text{Chi}^2 = 0.20$, $\text{df} = 1$ ($P = 0.65$), $I^2 = 0\%$), control groups (Analysis 1.12 in [Supplementary material 6](#); test for subgroup differences: $\text{Chi}^2 = 3.88$, $\text{df} = 4$ ($P = 0.42$), $I^2 = 0\%$), or published and recalculated data (Analysis 1.13 in [Supplementary material 6](#); test for subgroup differences: $\text{Chi}^2 = 0.05$, $\text{df} = 1$ ($P = 0.82$), $I^2 = 0\%$).

Serious adverse events

We did not do subgroup analyses of serious adverse events due to insufficient data.

Sensitivity analysis

All-cause mortality

The result of the sensitivity analysis using a fixed-effect model for all-cause mortality (RR 0.89, 95% CI 0.74 to 1.09) was compatible with the main analysis result (RR 0.89, 95% CI 0.71 to 1.12). A sensitivity analysis limited to trials at low risk of bias (RR 1.13, 95% CI 0.57 to 2.22; $I^2 = 0\%$; 5 studies, 275 participants; Analysis 1.14 in [Supplementary material 6](#)) shows a comparable result to the primary analysis that BCAAs may have little to no effect on all-cause mortality. All included trials were published as full papers, so analysis of publication status is equal to Analysis 1.1. Also, when we excluded trials with a lactulose or neomycin control group (RR 0.99, 95% CI 0.77 to 1.26; $I^2 = 0\%$; 11 studies, 574 participants; Analysis 1.15 in [Supplementary material 6](#)) or when the analysis only included trials with a lactulose or neomycin control (RR 0.76, 95% CI 0.43 to 1.32; $I^2 = 0\%$; 5 studies, 271 participants; Analysis 1.16 in [Supplementary material 6](#)), the results were compatible. None of the participants died in the trial that included people with or without cirrhosis (Calvey 1985), so exclusion of the trial did not influence the overall result (Analysis 1.17 in [Supplementary material 6](#)).

Hepatic encephalopathy

The result of the sensitivity analysis using a fixed-effect model for hepatic encephalopathy (RR 0.82, 95% CI 0.73 to 0.91) was compatible with the main analysis result (RR 0.79, 95% CI 0.64 to 0.96). The effect of BCAAs on hepatic encephalopathy was comparable to the respective overall analysis when we investigated trials with a low risk of bias (Analysis 1.18 in [Supplementary material 6](#)) or publication status (Analysis 1.19 in [Supplementary material 6](#); test for subgroup differences: $\text{Chi}^2 = 1.72$, $\text{df} = 1$ ($P = 0.19$), $I^2 = 41.9\%$). When excluding trials with a lactulose or neomycin control group, BCAAs may reduce hepatic encephalopathy (RR 0.80, 95% CI 0.63 to 1.00; $I^2 = 46\%$; 12 studies, 641 participants; Analysis 1.20 in [Supplementary material 6](#)). When the analysis only includes trials with a lactulose or neomycin control (RR 0.85, 95% CI 0.40 to 1.81; $I^2 = 12\%$; 5 studies, 271 participants; Analysis 1.21 in [Supplementary material 6](#)), the evidence is uncertain. In sensitivity analyses restricted to participants with cirrhosis, BCAAs showed effects comparable to those observed in the overall analysis (Analysis 1.22 in [Supplementary material 6](#)).

Serious adverse events

The result of the sensitivity analysis using a fixed-effect model for serious adverse events (nausea and diarrhoea) (RR 1.79, 95% CI 0.98 to 3.28) was compatible with the main analysis result (RR 2.05, 95% CI 0.40 to 10.58).

Summary of findings

We present key results for the outcomes of all-cause mortality, hepatic encephalopathy, serious adverse events, quality of life, and nutritional status outcomes in [Summary of findings 1](#), together with our GRADE assessments of the evidence certainty. We presented the results of the primary meta-analysis for our outcomes. All control groups were combined. As the table shows, we deemed the certainty of the evidence to be low or very low for all outcome measures, due to the high risk of bias in the trial results, indirectness, imprecision, and likelihood for publication bias.

Equity assessment

Not planned.

Reporting biases

One trial published as an abstract did not report all-cause mortality (Hayashi 1991). We had access to the protocols of four trials (Les 2011; Marchesini 1990; Marchesini 2003; Vilstrup 1990). All four reported pre-defined outcomes, such as mortality and hepatic encephalopathy, and were considered to be at low risk of bias in selection of the reported results. In trials with no protocol available, we compared the aims and methods in the Methods section of the article with text in the reported Results and Discussion sections. We found no discrepancies (Calvey 1985; Cerra 1985; Egberts 1985; Fiaccadori 1985; Hayashi 1991; Horst 1984; Hwang 1988; Mehtani 2023; Michel 1985; Muto 2005; Plauth 1993; Rossi-Fanelli 1986; Strauss 1986; Vidot 2019).

We found that 17 trials pre-defined and reported clinically relevant outcomes (mortality and hepatic encephalopathy). We judged these 17 trials to be at low risk of selective reporting bias (Calvey 1985; Cerra 1985; Egberts 1985; Fiaccadori 1985; Horst 1984; Hwang 1988; Les 2011; Marchesini 1990; Marchesini 2003; Mehtani 2023; Michel 1985; Muto 2005; Plauth 1993; Rossi-Fanelli 1986; Strauss 1986; Vidot 2019; Vilstrup 1990).

The funnel plots for mortality (Figure 3) and hepatic encephalopathy (Figure 4) appeared generally symmetrical, with trials evenly distributed around the mean effect size, suggesting a low likelihood of publication bias.

DISCUSSION

Summary of main results

This updated review added two new trials. We thus included a total of 18 randomised clinical trials published from 1984 to 2023. With the addition of two trials to this review update, we still did not find any beneficial or harmful effect of branched-chain amino acids (BCAAs) on all-cause mortality. Results showed a narrower confidence interval.

Evidence suggests that BCAAs reduce the manifestations of hepatic encephalopathy in people with cirrhosis. The trials compared BCAAs to placebo or no intervention, diets, lactulose, or neomycin. The certainty of the evidence is low due to risk of bias. The between-trial heterogeneity was moderate.

We are uncertain whether BCAAs, compared with control, have any effect on all-cause mortality, nausea, diarrhoea, serum albumin, or nitrogen balance, as the certainty of the evidence was generally very low. Data on quality of life could not be meta-analysed. We did not identify any trials in children. We found no randomised clinical trials comparing BCAAs with rifaximin.

Limitations of the evidence included in the review

When evaluating the quality of the included trials, both internal and external validity were important. Internal validity must be sufficient before considering external validity, as inadequate internal validity renders external validity irrelevant. Without adequate external validity, a trial may be clinically irrelevant despite having excellent basic methodology. To ensure internal validity, it is essential to control for bias associated with blinding and randomisation, as

these factors can lead to overestimated intervention effects [152, 153]. When evaluating bias, we estimated that the lack of blinding would not affect the assessment of all-cause mortality or laboratory parameters. In a study including 1973 randomised clinical trials from 234 meta-analyses [153], authors found that trials with inadequate or unclear (as opposed to adequate) sequence generation or allocation concealment overestimated intervention effects. The degree of bias was greatest for subjective outcomes, and there was little evidence of bias for all-cause mortality (relative odds ratio 0.89; credibility interval 0.75 to 1.05). Lack of, or unclear, double-blinding (as opposed to double-blinding) was associated with a mean 13% exaggeration of intervention effects. Again, bias was detected in analyses of subjective outcomes and not for all-cause mortality. When we included only the trials at low risk of bias, we found no effect of BCAAs on all-cause mortality, but we found a beneficial effect on hepatic encephalopathy.

The benefits of the cross-over study design include comparable control groups and increased statistical power. However, the use of the design in hepatic encephalopathy is questionable because the condition is a naturally fluctuating disease and because people may die during the first treatment period [36, 116]. We chose to use only data from the first period of the two cross-over trials included in the review despite the potential loss of information that this strategy entails [154]. The duration of therapy ranged from four days to 104 weeks. Because we used the first treatment period data from the cross-over trials, the information that we included from these trials was based on a relatively shorter follow-up than trials using a parallel-group design. In our subgroup analyses, we found no difference between cross-over and parallel-group trials. Yet, we cannot rule out the possibility that the duration of follow-up and trial design may be significant factors.

Previous guidelines recommended oral BCAAs as an alternative or add-on therapy to the first-line treatment, which is non-absorbable disaccharides [7, 39]. In the latest guidelines from the European Association for the Study of the Liver (EASL), rifaximine has replaced BCAAs as an adjunct to non-absorbable disaccharides [8]. Our updated review continues to support the use of BCAAs in clinical practice, but it does not provide sufficient evidence to evaluate the benefit of BCAAs compared to other interventions.

The cost-effectiveness of interventions for people with hepatic encephalopathy is important. During the first year, the incremental adjusted cost of caring for people with cirrhosis in the USA was USD (US dollars) 35,029 (95% confidence interval USD 32,473 to USD 37,585) [155]. Management of an individual with hepatic encephalopathy during the same period was USD 50,080, being more expensive than varices and hepatocellular cancer [155]. We did not assess the costs associated with hepatic encephalopathy, but our data may be used in future pharmacoeconomic reviews.

The included trials were heterogeneous. We decided to keep the broad scope of the review to identify common features within similar interventions. Grouping increases the external validity and consistency of the findings, enabling us to assess the intervention effect across a wider range of settings and patient populations. The strategy also reduces chance results that can occur when analysing several smaller subgroups [156]. In theory, grouping makes it possible to make better judgements about the consistency of observed effects across trials and to look for differences between interventions, settings, and participant characteristics. We did not identify predictors of between-trial heterogeneity. None

of the intervention or participant characteristics explained the heterogeneity. However, we did not have sufficient data to exclude the possibility that specific patient populations might benefit more or less from BCAAs.

Most of the participants had cirrhosis. Calvey 1985 included 64 people with acute liver disease with or without cirrhosis. Of these 64 participants, 30 had cirrhosis, and we included only 26 people with overt hepatic encephalopathy at baseline in our analyses. No one died in the Calvey 1985 trial, and the overall mortality analysis was not affected by its inclusion. When hepatic encephalopathy occurs, severe liver dysfunction must be present. Yet, the distribution of cirrhosis in the 26 participants was unknown.

The diagnostic criteria used in individual trials varied. The variation largely reflected the differences that occurred between the publication of the first trials in 1984 (Fiaccadori 1985; Horst 1984), and the trial from 2005 (Muto 2005). Using a systematic approach for assessment of hepatic encephalopathy was essential [4]. The included trials used methods such as the West Haven Criteria [14, 157, 158]. This system grades symptoms from a clinically normal mental state (grade 0), to trivial signs such as shortened attention span, to deep coma (grade 4). The clinically normal state includes people without overt hepatic encephalopathy and people with minimal hepatic encephalopathy that can be identified through psychometric testing, such as the number connection test [4, 14]. In one cohort study of 217 people with cirrhosis, repeated evaluations showed that hepatic encephalopathy was not classed correctly in more than one-third of people [159]. Therefore, thorough assessments are important. Likewise, careful evaluation is required to evaluate whether manifestations of hepatic encephalopathy are improved. Based on the differences in the diagnostic criteria, the assessment of this outcome measure varied in included trials. We chose to use the information and evaluation made by the original investigators. The extent to which manifestations improve will depend on the pre-defined criteria, and the overall assessment is likely to involve some degree of subjectivity. Despite these limitations, the definition and outcome assessment reflect current clinical practice [160].

One study found that BCAA supplementation increased ammonia levels in people with cirrhosis as well as healthy participants [25]. BCAAs stimulate ammonia detoxification by promoting glutamine synthesis and are decreased in liver cirrhosis [26]. The reasons for the BCAA deficiency in cirrhosis include consumption in skeletal muscle for the synthesis of glutamate. We were unable to determine the effect of BCAAs on the baseline BCAA concentrations, sarcopenia, or ammonia levels.

No trials included children, and our conclusions can only be applied to adults.

We were unable to conduct sensitivity analyses to evaluate the impact of missing outcome data [161]. Although many trials appeared to report complete follow-up data, this was not stated explicitly, and we were therefore unable to verify the absence of missing data. Four trials reported losses to follow-up but did not provide the specific information required to calculate attrition or to apply best-worst and worst-best case analyses (Fiaccadori 1985; Hayashi 1991; Rossi-Fanelli 1986; Vilstrup 1990). Owing to the uncertainty introduced by incompletely reported attrition, we considered the risk of attrition bias to be potentially important.

Consequently, we judged these trials to be at high risk of bias, and downgraded the certainty of the evidence.

Limitations of the review processes

The main potential bias in the review process is that we only had access to some trials' published data. We found no difference between the subgroups of trials with published summary data and trials for which we recalculated outcomes for people with hepatic encephalopathy at baseline. On the one hand, this supports the strength of our findings. We cannot exclude that we would have been able to retrieve additional information if we had access to unpublished and published data from all trials. One of the included trials was only published in abstract form [104]. In some trials, the complete results presented in the final full-text article differed from the initial analyses presented in abstract form [162]. Access to the original patient data from trials published in abstract form may increase the strength of the analyses. Due to the risk of publication bias, we did not exclude trials published in abstract form to avoid overestimation of intervention benefits.

We planned to use Peto's odds ratio for our analysis of adverse events. The Peto odds ratio is most reliable when treatment effects are small and events are rare. Our included trials varied in participant populations, BCAA administration routes (oral versus intravenous), and dosing regimens, causing heterogeneity. In this case, we found using a fixed-effect meta-analysis as sensitivity analysis more appropriate.

Whether BCAAs prevent recurrences of hepatic encephalopathy is still a question to be addressed.

Agreements and disagreements with other studies or reviews

For this review update, the overall study findings and conclusions were consistent with those of our prior review. The previous version of this review included 16 randomised clinical trials with 827 participants [47]. This updated review included 18 trials with 934 participants. Both versions compared trials using oral or intravenous BCAAs to any control intervention. In both reviews, BCAAs were not associated with reduced all-cause mortality. Likewise, both reviews found that BCAAs were associated with a beneficial effect on hepatic encephalopathy. In the previous review, an analysis limited to trials at a low risk of bias confirmed that BCAAs have a beneficial effect on hepatic encephalopathy. Accordingly, the overall quantitative results of the two reviews are consistent.

One comprehensive Cochrane review by Koretz and colleagues from 2012 included randomised clinical trials on any type of nutritional support for any type of liver disease [163]. The review found no difference in all-cause mortality between nutritional therapy versus no nutritional therapy for liver disease in a medical setting. One meta-analysis of six trials (total 119 participants) found that nutritional supplements had a beneficial effect on resolution of hepatic encephalopathy. Two of the six trials evaluated BCAAs (Calvey 1985; Hayashi 1991). One trial included people with acute alcoholic hepatitis (Calvey 1985), and found no differences between BCAAs and control groups regarding all-cause mortality, hepatic encephalopathy, or nutritional parameters. The second trial included people with cirrhosis, and found a beneficial effect of BCAAs on improvement of hepatic encephalopathy (Hayashi 1991).

The Koretz systematic review found that the effect of nutritional supplements on hepatic encephalopathy was more pronounced in subgroup analyses of trials using a high concentration of BCAAs [163]. However, there was unclear control of bias, the total number of participants was small, and the overall result was still consistent with our findings.

Recent meta-analyses have examined BCAAs' effect on people with cirrhosis, the overall complications (+/- hepatic encephalopathy), and all-cause mortality [164, 165, 166, 167]. Two analyses, which included nine studies (total 1080 participants) [164] and 20 studies (total 1298 participants) [166], respectively, concluded that BCAAs reduced the incidence of serious complications. Only the smaller of the two meta-analyses by Du and colleagues included studies exploring hepatic encephalopathy as a complication [164]. Two other meta-analyses concluded (a) that BCAAs were safe [165], and (b) could improve prognosis in cirrhosis [167], respectively. The meta-analysis by Van Dijk and colleagues included 54 studies with 5184 participants [165], and the one by Lee and colleagues, 28 studies with 3305 participants [167]. Only the latter meta-analysis specifically studied hepatic encephalopathy in a subgroup analysis (seven studies with 550 participants), and concluded that BCAAs prevented hepatic encephalopathy [167].

AUTHORS' CONCLUSIONS

Implications for practice

This review found that branched-chain amino acids (BCAAs) may have little to no effect on all-cause mortality, but the certainty of the evidence is very low. The evidence suggests that BCAAs could be considered in the management of people with hepatic encephalopathy as the manifestations of hepatic encephalopathy associated with cirrhosis may be reduced, but the certainty of the evidence is low. Adverse events mainly included gastrointestinal symptoms. BCAAs may have little to no effect on nutritional outcomes, but the certainty of the evidence is very low. Quality of life could not be meta-analysed.

Implications for research

We need additional high-quality trials to evaluate the effect of BCAAs combined with interventions that are used in clinical practice, such as antibiotics and lactulose [7, 39]. We also need trials in people with minimal hepatic encephalopathy and trials with data on participants' quality of life and nutritional parameters. Additional research is also needed to evaluate whether BCAAs have a beneficial effect on hepatic encephalopathy in people with acute liver failure without cirrhosis and to identify if specific subgroups of people benefit from treatment with BCAAs.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD001939.pub5](https://doi.org/10.1002/14651858.CD001939.pub5).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of ongoing studies

Supplementary material 5 Risk of bias

Supplementary material 6 Analyses

Supplementary material 7 Data package

Supplementary material 8 Differences between the protocol and previous versions

ADDITIONAL INFORMATION

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Editorial and peer-reviewer contributions

Cochrane Hepato-Biliary Group supported the authors in the development of this review update.

The following people from the Cochrane Hepato-Biliary Editorial Team conducted the internal editorial process for this review:

- Sign-off Editor (final editorial decision): Christian Gluud, Denmark;
- Managing Editor (provided editorial guidance to authors, edited the review): Dimitrinka Nikolova, Denmark;
- Information Specialist (search strategy design): Sarah Louise Klingenberg, Denmark.

The following people conducted the editorial process for the 2025 review update:

- Sign-off Editor (final editorial decision): Agostino Colli MD, the Cochrane Hepato-Biliary Group, Copenhagen Trial Unit, Centre for Clinical Intervention Research, The Capital Region, Copenhagen University Hospital - Rigshospitalet, Copenhagen, Denmark;
- Managing Editor (provided editorial guidance to authors, edited the article): Anupa Shah, Cochrane Central Editorial Service;
- Editorial Assistant (conducted editorial policy checks, selected peer reviewers, collated peer-reviewer comments and supported the editorial team): Andrew Savage, Cochrane Central Editorial Service;
- Copy Editor (copy editing and production): Faith Armitage, Cochrane Central Production Service;
- Peer-reviewers (provided comments and recommended an editorial decision): Nicolas Weiss, Sorbonne Université, La Pitié-Salpêtrière, Paris (clinical/content review); Clare Miles, Evidence

Production and Methods Directorate (methods review); Jo Platt, Central Editorial Information Specialist (search review).

Contributions of authors

LA, ND, GD, and LLG reviewed and updated the methods based on the previous review, with incorporation of the revised guidelines described in the *Cochrane Handbook for Systematic Reviews of Interventions*.

IL, GM, MB, NKA, and HV made substantial contributions to the review.

All the authors approved the final version of the manuscript.

Declarations of interest

None of the authors has any financial interests that may conflict with this review.

Luise Aamann: none known

Neha Despande: none known

Gitte Dam: none known

Iñigo Les was an investigator of an included randomised clinical trial (Les 2011). Iñigo Les was not involved with the selection, data extraction, risk of bias, or GRADE assessment of his trial in this review. Neither was Iñigo Les involved in the selection, data extraction, risk of bias, or GRADE assessment of the other three trials included in this review, with investigators Giulio Marchesini and Hendrik Vilstrup (Marchesini 1990; Vilstrup 1990; Marchesini 2003).

Giulio Marchesini was an investigator of two included randomised clinical trials (Marchesini 1990; Marchesini 2003). Giulio Marchesini was not involved with the selection, data extraction, risk of bias, or GRADE assessment of his trials in this review. Neither was Giulio Marchesini involved in the selection, data extraction, risk of bias, or GRADE assessment of the other two trials included in this review, with investigators Hendrik Vilstrup and Iñigo Les (Vilstrup 1990; Les 2011).

Mette Borre: none known

Niels Kristian Aagaard: none known

Hendrik Vilstrup was an investigator of an included randomised clinical trial (Vilstrup 1990). Hendrik Vilstrup was not involved with the selection, data extraction, risk of bias, or GRADE assessment of his trial in this review. Neither was Hendrik Vilstrup involved in the selection, data extraction, risk of bias, or GRADE assessment of the other three trials included in this review, with investigators Giulio

Marchesini and Hendrik Vilstrup (Marchesini 1990; Marchesini 2003; Les 2011).

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Registration and protocol

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Review update (2017): doi.org/10.1002/14651858.CD001939.pub4 [47]

Data, code and other materials

As part of the published Cochrane review, the following is made available for download for users of the Cochrane Library (Supplementary material 7): full search strategies for each database; full citations of each unique report for all studies included, ongoing or waiting classification, or excluded at the full text screen, in the final review; study data, including study information, study arms, and study results or test data; consensus risk of bias assessments; and analysis data, including overall estimates and settings, subgroup estimates, and individual data rows. Appropriate permissions have been obtained for such use. Analyses and data management were conducted within Cochrane's authoring tool, RevMan, using the inbuilt computation methods. Template data extraction forms from Excel are available from the authors on reasonable request.

What's new

Date	Event	Description
16 January 2026	New citation required but conclusions have not changed	Two new trials added (in total 107 participants), but the overall conclusions remained the same
16 January 2026	New search has been performed	New search

Date	Event	Description
		- Updated the risk of bias evaluation from the Cochrane Hepato-Biliary Group method to version 2 (RoB 2) - This review is now presented in a focused article format. - The statistical methods were updated with those methods currently available and recommended for use in RevMan, including the Restricted Maximum Likelihood (REML) estimator and the Hartung-Knapp-Sidik-Jonkman (HKSJ) method. - The certainty of the evidence was downgraded to either low or very low certainty. - Two new authors have been added (LA and ND).

History

Protocol first published: Issue 1, 1997

Review first published: Issue 2, 2003

Date	Event	Description
4 March 2020	Amended	Clarification message from the Co-ordinating Editor added to the Declarations of interest statement about the review's compliance with the Cochrane conflict of interest policy, which includes the relevant parts of the Cochrane Commercial Sponsorship Policy.
5 May 2017	New search has been performed	New search conducted.
5 May 2017	New citation required but conclusions have not changed	No additional trials identified.

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ADDITIONAL TABLES

Table 1. Overview of synthesis and included studies (OSIS) table

Study name (year) country	Study designs	Intervention	BCAA details	Participants: sample size = n (BCAAs/control)	Proportion	Outcomes with available data; synthesis method for all listed outcomes was meta-analysis	Outcome measures	Outcome time point included in MA
				Type of hepatic encephalopathy				
				Mean age (BCAAs/control)				
Calvey 1985 UK	Parallel-arm randomised clinical trial	BCAAs versus non-isonitrogenous diet	Oral or intravenous (people who deteriorated) BCAAs 20 g/day for 3 weeks	- n = 26 (10/16) - Overt 49/49 years (median)	- Cirrhosis 47% - ALD 100% - VH 0% - Men 48%	- Mortality - Hepatic encephalopathy	Hepatic encephalopathy: - Modified 0- to 5-point Glasgow Coma Score - Reitan Trail-Making Test	3 weeks
Cerra 1985 USA	Parallel-arm randomised clinical trial	BCAAs versus neomycin	Intravenous BCAAs (36% BCAA solution 1.5 to 3 L/day) 4 days	- n = 75 (40/35) - Overt 53/53 years	- Cirrhosis 92-100% - ALD 92% - VH 0% - Men 86%	- Mortality - Hepatic encephalopathy - Nitrogen balance	Hepatic encephalopathy: 0- to 4-point score	4 days
Egberts 1985 Germany	Cross-over randomised clinical trial	BCAAs versus isonitrogenous diet	Oral BCAAs (0.25 g/kg/day) for 7 days	- n = 22 (11/11) - Minimal 51/52 years	- Cirrhosis 100% - ALD 86% - VH 14% - Men 73%	- Mortality - Hepatic encephalopathy - Nitrogen balance	Hepatic encephalopathy: 10 different tests, e.g. - Motor Performance Test Battery - Vienna Reaction Time Apparatus	7 days
Fiaccadori 1985 Italy (Abstract)	Parallel-arm randomised clinical trial	BCAAs alone or with lactulose versus lactulose alone	Intravenous BCAAs (2 L/day) for 7 days	- n = 48 (32/16) - Overt 51/51 years	- Cirrhosis 100% - ALD 52% - VH 32% - Men 73%	- Mortality - Hepatic encephalopathy	Hepatic encephalopathy: - Clinical evaluations - Asterixis - Blood ammonia - Electroencephalography - Reitan Number Connection Test	7 days

Table 1. Overview of synthesis and included studies (OSIS) table (Continued)

Hayashi 1991 Japan (Abstract)	Parallel-arm randomised clinical trial	BCAAs versus isonitrogenous diet	Oral BCAAs (11 g/day) for 3 weeks	• n = 67 (35/32) • Overt • age not reported	• Cirrhosis 100% • ALD 58% • VH 9% • Men 68%	• Hepatic encephalopathy • Nausea and diarrhoea	Hepatic encephalopathy: 3 weeks	• 4-point score • Number Connection Test • Serial Seven Test
Horst 1984 USA	Parallel-arm randomised clinical trial	BCAAs versus isonitrogenous diet	Oral BCAAs 20 g/day increased to 80 g/day for 4 weeks	• n = 37 (17/20) • Overt • 57/60 years	• Cirrhosis 100% • ALD 38% • VH 8% • Men 65%	• Mortality • Hepatic encephalopathy • Nausea and diarrhoea	Hepatic encephalopathy: 4 weeks	• 4-point score • Number Connection Test • Electroencephalography
Hwang 1988 China	Parallel-arm randomised clinical trial	BCAAs versus no intervention	Intravenous BCAAs 40 g/day for 5 days	• n = 55 (27/28) • Overt • 60/62 years	• Cirrhosis 100% • ALD 15% • VH 65% • Men 90%	• Mortality • Hepatic encephalopathy	Hepatic encephalopathy 5 days	• 0- to 4-point scale
Les 2011 Spain	Parallel-arm randomised clinical trial	BCAAs versus isocaloric diet	Enteral BCAAs 29 g/day versus isocaloric placebo for 56 weeks	• n = 40 (18/22) • Minimal • 64/63 years	• Cirrhosis 100% • ALD 36% • VH 53% • Men 76%	• Mortality • Hepatic encephalopathy • Albumin	Hepatic encephalopathy 56 weeks	• 4-point score • Trail Making Test part A • Symbol Digit Test (oral version) • Grooved Pegboard Test (dominant hand)
Marchesini 1990 Italy	Parallel-arm randomised clinical trial	BCAAs versus isonitrogenous control versus isocaloric casein-based control	Oral BCAAs 2.4 g/10 kg/day for 3 months.	• n = 64 (30/34) • Overt • 60/60 years	• Cirrhosis 100% • ALD 56% • VH 41% • Men 80%	• Mortality • Hepatic encephalopathy • Albumin • Nitrogen balance	Hepatic encephalopathy 3 months	• 12-point score
Marchesini 2003 Italy	Parallel-arm randomised clinical trial	BCAAs versus isonitrogenous and isocaloric casein-based control	Oral BCAAs 7.2 g/day for 12 months.	• n = 112 (33/79) • Minimal • 59/60 years	• Cirrhosis 100% • ALD 21% • VH 68% • Men 63%	• Mortality • Hepatic encephalopathy • Nausea and diarrhoea • Albumin	Hepatic encephalopathy 12 months	• 4-point score • Reitan Test

Table 1. Overview of synthesis and included studies (OSIS) table (Continued)

Mehtani 2023 India	Parallel-arm randomised clinical trial	BCAAs and lactulose versus lactulose	Intravenous BCAAs 500 ml/day (+ Lactulose 30-60 ml x 3/day) for 3 days	- n = 76 (39/37) - Overt 47/49 years	- Cirrhosis 100% - ALD 74% - VH 14% - Men 90%	- Mortality - Hepatic encephalopathy - Nausea and diarrhoea	Hepatic encephalopathy: HESA	Mortality: 28 days Hepatic encephalopathy: 7 days
Michel 1985 France	Parallel-arm randomised clinical trial	BCAAs versus isonitrogenous and isocaloric control	Intravenous BCAAs 6.25 to 12.5 g/day for 5 days	- n = 70 (36/34) - Overt 61/59 years	- Cirrhosis 100% - ALD 81% - VH 11% - Men 30%	- Mortality - Hepatic encephalopathy	Hepatic encephalopathy: 3-point scale	36 days (1 month after treatment)
Muto 2005 Japan	Parallel-arm randomised clinical trial	BCAAs versus non-isonitrogenous control diet	Oral BCAAs 12 g/day for 2 years	- n = 39 (27/12) - Overt 62/61 years	- Cirrhosis 100% - ALD 8% - VH 81% - Men 47%	- Mortality - Hepatic encephalopathy - Nausea and diarrhoea	Hepatic encephalopathy: 4-point scale	2 years
Plauth 1993	Cross-over randomised clinical trial	BCAAs versus placebo	Oral BCAAs 0.25 g/kg/day for 8 weeks	- n = 23 (12/11) - Minimal 52/49 years	- Cirrhosis 100% - ALD 88% - VH 12% - Men 65%	- Mortality - Hepatic encephalopathy - Nausea and diarrhoea	Hepatic encephalopathy: - Digit Symbol - Number Connection Test - Test batteries on psychomotor function, attention, and reaction time	8 weeks
Rossi-Faneli 1986 Italy	Parallel-arm randomised clinical trial	BCAAs versus lactulose	Intravenous BCAAs (57 g/day) for 2 days	- n = 40 (20/20) - Overt 59/59 years	- Cirrhosis 59% - ALD 32% - VH 12% - Men 62%	- Mortality - Hepatic encephalopathy	Hepatic encephalopathy: 4-point scale	2 days
Strauss 1986 Brazil	Parallel-arm randomised clinical trial	BCAAs versus neomycin	Intravenous BCAAs (1.2 to 1.5 g/kg/day) for maximum 5 days	- n = 32 (16/16) - Overt 51/54 years	- Cirrhosis 100% - ALD 75% - VH 13% - Men 90%	- Mortality - Hepatic encephalopathy	Hepatic encephalopathy: 4-point scale	5 days

Table 1. Overview of synthesis and included studies (OSIS) table *(Continued)*

Vidot 2019 Australia	Parallel-arm randomised clinical trial	BCAAs versus Synbiotics ver- sus a combina- tion of synbi- otics and BCAAs versus placebo	Oral BCAAs (BCAAs 6 g/day) for 8 weeks	• n = 31 (15/16) • Minimal • 56/55 years	• Cirrhosis 100% • ALD 23% • VH 48% • Men 85%	• Mortality • Hepatic en- cephalopathy	Hepatic encephalopathy: • Inhibitory Control Test • Trail-Making Test	8 weeks
Vilstrup 1990 Denmark	Parallel-arm randomised clinical trial	BCAAs versus isonitrocaloric control	BCAAs (12.5 mL/kg/day) for maximum 16 days	• n = 77 (38/39) • Overt • 55/56 years	• Cirrhosis 100% • ALD 91% • VH 9% • Men 72%	• Mortality • Hepatic en- cephalopathy	Hepatic encephalopathy: • 4-point scale	16 days

ALD: alcohol-associated liver disease; **BCAAs:** branched-chain amino acids; **HESA:** Hepatic Encephalopathy Scoring Algorithm; **VH:** viral hepatitis